

Minji Shon

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Work Authorization: F-1, CPT/OPT eligible | EB-2 NIW I-140 approved — no employer sponsorship needed for green card

Summary

Ph.D. candidate in ECE at Georgia Tech (expected **December 2027**) researching **device-to-system electro-thermal and reliability co-optimization**, with **7 years at Samsung Electronics** delivering aging-aware design verification and reliability sign-off for Exynos SoCs with FinFET technology. **Pre-silicon**: aging, safe operating area, and self-heating sign-off methodology; statistical aging simulation for automotive qualification. **Post-silicon**: model-to-hardware correlation against ring-oscillator and product silicon.

Education

Georgia Institute of Technology  — Atlanta, GA, USA

Aug 2023–Dec 2027
(expected)

Ph.D., Electrical and Computer Engineering

Sogang University  — Seoul, South Korea

Mar 2012–Aug 2016

B.S., Electronic Engineering; **summa cum laude**

Research Experience

Graduate Research Assistant — Georgia Institute of Technology , GA, USA

Aug 2023–Aug 2026

SRC Research Scholar (SPARC) — Advisor: Dr. Shimeng Yu

Ph.D. thesis: *Next-Generation 3D Scaling: Reliability-Aware Device Modeling and Design Co-Optimization for Advanced CMOS Logic and Memory*.

- **Electro-thermal reliability of 3D integration** — device-to-circuit framework (TCAD → self-heating-aware compact model → ring-oscillator analysis) showing that **backside power delivery** traps heat and costs **>10% frequency degradation** at overdrive — directly relevant to overclocked AI/HPC applications. [R1, R2]
- **Layout-resolved thermal analysis at scale** — GPU-accelerated multi-physics extended to multi-tier CFET, moving self-heating analysis from single-device TCAD to full-layout turnaround. [R1]
- **Ferroelectric memory reliability** — time-dependent Sentaurus models of trap/polarization interplay in FeFET and Fe-NAND stacks, spanning gate-stack and channel-side interlayer engineering [F1–F4]
- **Reliability-aware DTCO** — TCAD-calibrated device models and an **open-source CFET PDK** (six 67-cell library variants) taken through full RTL-to-GDS in commercial EDA; **40% logic** and **43% SRAM** cell-footprint reduction from 2-tier to 4-tier, giving reliability sign-off a path into early technology definition. [L1–L3, T1]

Industry Experience

Design Quality & Reliability Engineer Intern — Intel , Hillsboro, OR, USA

Aug 2026–Jan 2027

- Product Quality Network team: developing **physics-informed machine learning** methods for **pre-silicon reliability** of analog IP and memory circuits, targeting CPU, server, and ASIC products.

Technology Quality & Reliability Engineer — Samsung Electronics , Korea

Aug 2016–Jul 2023

Pre-silicon reliability sign-off & aging-aware design verification

- Delivered aging- and variability-aware design methodologies for **Samsung Exynos SoCs (14 nm–4 nm)** under advanced reliability sign-off.
- Developed **statistical aging simulation modeling and methodology**, enabling **AEC-Q100** automotive-grade design-for-reliability in **8 nm FinFET** products.
- Partnered with **Design Technology Enablement** to deliver **critical-path reliability timing margin** from aging simulation directly into the Exynos SoC implementation flow.
- Built aging-aware circuit design environments and **PDK components** for **3 nm GAA nanosheet** technology.

Post-silicon correlation & reliability model validation

- Built and validated reliability models for **HCI, BTI, TDDB, and self-heating**, improving prediction accuracy through **model-to-hardware correlation** against **ring-oscillator** silicon benchmarks with the WLR and PDK teams.
- Provided DFR for **real product operating conditions** (multi-step aging, power-down bias instability) rather than idealized stress, closing the gap between qualification and field use. [R3]

IP- & memory-level reliability verification

- Verified **overdrive-voltage reliability** in non-volatile memory (**eFuse, OTP, MRAM**) under product stress conditions.

- *Prior internship (Jul–Aug 2016)*: implemented **enhanced Dynamic Device Voltage Check (DDVC)**; strengthened **safe-operating-area (SOA)** validation and **voltage-stress reliability sign-off** across advanced nodes.

Technical Skills

Reliability physics & sign-off: HCI, BTI, TDDb, self-heating, SOA; statistical aging simulation; model-to-hardware correlation; AEC-Q100 automotive qualification.

Circuit & reliability simulation: SPICE (HSPICE, FineSim, Spectre) with aging engines (**RelXpert**, **MOSRA**); Dynamic Device Voltage Check (DDVC); multi-step and power-down-bias stress modeling.

Pre-silicon flows: PDK build & verification (LIB/LEF, LPE, DRC/LVS); Synopsys/Cadence synthesis, P&R and reliability sign-off; critical-path timing-margin analysis.

TCAD & device modeling: Sentaurus electro-thermal device modeling; BSIM-CMG compact modeling.

Technologies: FinFET, GAA nanosheet, multi-tier CFET, backside power delivery, ferroelectric-based memory.

Programming & HDL: Python, TCL, Perl, C, MATLAB, SystemVerilog.

Publications (Selected, by Topic)

* denotes equal contribution. Five papers at *IEEE IRPS* (2019, 2025, 2026×2, 2027).

Reliability & Electro-Thermal Co-Design

- [R1] **Shon, M.***, McClellan, C. J.*, et al., Yu, S. (2026). *AI-Accelerated, Layout-Resolved Electro-Thermal Analysis for Multi-Tier CFET Logic*. IRPS'27 (submitted).
- [R2] Dutta, R.*, **Shon, M.***, et al., Datta, S. (2026). *Self-Heating in GAA Nanosheet FETs with integrated Back Side Power Delivery Network: Device to Circuit Electro-Thermal Analysis and Implications*. IRPS'26. [DOI](#)
- [R3] Shim, H., Jo, J., Kim, Y., Jeong, B., **Shon, M.**, et al., Pae, S. (2019). *Aging-Aware Design Verification Methods Under Real Product Operating Conditions*. IRPS'19. [DOI](#)

Ferroelectric Device & Memory Reliability

- [F1] Fernandes, L.*, **Shon, M.***, et al., Khan, A. (2026). *Channel-Side Interlayer Engineering in Laminated FeFETs: Trap-Driven Optimization of Memory Window, Retention, and Disturb in Ferroelectric NAND Flash*. IRPS'26. [DOI](#)
- [F2] Phadke, O., **Shon, M.**, et al., Yu, S. (2025). *On the Localized Ferroelectric Phase Variation in Scaled FeFET: Experiments and Modeling*. IEDM'25. [DOI](#)
- [F3] **Shon, M.**, et al., Yu, S. (2025). *Modeling Dynamic Interplay Between Charge Traps and Polarization for Memory Window Enhancement in Gate Injection Layers*. IRPS'25. [DOI](#)
- [F4] **Shon, M.**, et al., Yu, S. (2025). *A Comprehensive Modeling of Gate Stack Interlayer Engineering for Ferroelectric Vertical NAND*. DRC'25 (Best Paper Runner-Up). [DOI](#)

3D Logic & SRAM DTCO/STCO

- [L1] **Shon, M.***, Lim, S.*, et al., Yu, S. (2026). *MATRIX-A5: An Open-Source CFET PDK with CFlex Track-Mixing Libraries for RTL-to-GDSII DTCO*. JxCDC'26 (under review).
- [L2] **Shon, M.***, Lim, S.*, Yu, S. (2025). *DTCO of Multi-Tier CFETs: Ultimate Logic and SRAM Scaling of A5 Technology and Beyond*. IEDM'25. [DOI](#)
- [L3] Lim, S., **Shon, M.**, Yu, S. (2025). *3-Tier SRAM Complementary FET (CFET) High Density (HD) Cell Design at A5 Node*. TED'25. [DOI](#)
- [L4] Lee, J.*, **Shon, M.***, et al., Yu, S. (2025). *3D Digital Compute-in-Memory Benchmark with A5 CFET Technology: An Extension to Look-up-Table Based Design*. TVLSI'25. [DOI](#) (earlier version: APCCAS'24, [DOI](#))

Tutorial presentation

- [T1] *MATRIX Open-Source PDKs for Advanced DRAM and Sub-1-nm CFET Logic: From Device Models to RTL-to-GDSII Design-Technology Co-Optimization*. ASP-DAC'27. (Presenter: Yu, S, **Shon, M.**)

Honors & Awards

- **Best Paper Award, Runner-Up** — IEEE Device Research Conference (DRC) Jun 2025
- **Silicon Creations ISSCC Travel Grant** Feb 2024
- **Conference Travel Grants** — APCCAS'24, IRPS'25, DRC'25, IEDM'25 (presentation) 2024–2025
- **CSE Department Fellowship (offered)** — University of Minnesota–Twin Cities Jan 2023
- **Summa cum laude** — Sogang University; Dean's List Aug 2016
- **Full National Scholarship (Merit)** — Korea Student Aid Foundation 2014–2016
- **Honors Scholarship (Merit)** — Sogang University 2012–2015

Leadership & Service

- **Ministry Team Leader, Grad CRU** — Georgia Institute of Technology 2024–Present
- **Education Mentor / Sub-Course Manager** — Samsung Electronics 2019, 2020, 2022
 - Delivered onboarding for **200+** new hires; “Global Business Manners,” “Communication & Creative Thinking.”